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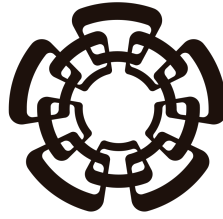
Sensibilidades a radio de carga de neutrinos en experimentos de neutrinos de reactor involucrando $CE\nu NS$

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Para la obtención del título de
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Centro de Investigación y de Estudios Avanzados del Instituto
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PHYSICS DEPARTMENT

**Sensitivities to neutrino charge radius at reactor
neutrino experiments involving $\text{CE}\nu\text{NS}$**

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Resumen

resumen aqui

Abstract

resumen en ingles

Contents

1	Introduction	1
2	Electroweak theory in the Standard Model of particles	2
2.1	Introduction	2
2.2	Symmetries and fermion content	2
2.2.1	Neutrinos	2
2.2.2	Weak Nuclear Force	4
2.3	Massive and Massless Mediators	6
2.3.1	Photons	6
2.3.2	$W^\pm$ and Z	7
2.4	Electroweak mixing and $\sin^2 \theta_W$	8
2.4.1	$SU(2)_L$	8
2.4.2	$SU(2)_L \times U(1)_Y$	10
3	Radiative Corrections and Neutrino Charge Radius	13
3.1	Introduction	13
3.2	Electromagnetic Form Factors	13
3.2.1	Massive Neutrinos	13
3.2.2	Massive Neutrino Mixing	16
3.2.3	Flavor Base	17
3.2.4	Weyl Neutrinos	18
3.3	Neutrino Charge Radius	18
3.4	Corrections to the $\nu\nu\gamma$ vertex	20
3.5	NCR as a radiative correction of Electroweak Mixing	21
4	Coherent Elastic Neutrino-Nucleus Scattering	22
4.1	Introduction	22
4.2	Cross Section	22
4.3	Neutrino Sources	23
4.3.1	Nuclear Reactors	23
4.4	Expected Number of Events	25
5	SBC-ININ-LAr	26
5.1	Results	26
6	Summary and conclusions	28
	Bibliography	30

1. Introduction

intro

2. Electroweak theory in the Standard Model of particles

This chapter serves as a description of the foundation for the theoretical framework of this work. Modern particle physics is based on the mathematical modeling of particles as excitations of quantum fields and how these fields interact with each other. This chapter begins with a brief history of the neutrino followed by its importance as the catalyst for the proposal and eventual discovery of a fundamental nuclear force, followed by its unification with a previously well understood force to introduce the main theory by which further advancements in neutrino property studies can be made.

2.1. Introduction

The Standard Model of particles (SM) is the most currently utilized theory of interactions of subatomic particles. It describes all matter particles and forces between them, excluding gravity, as interactions between 12 fermions (6 quarks and 6 leptons) with their corresponding anti-fermions, 4 spin-1 force-carrying gauge bosons and 1 scalar boson as shown in Fig. 2.1. Within this framework exist 3 distinct forces: strong nuclear force, mediated by the gluon (g) boson; electromagnetic (EM) force, mediated by the photon (γ) boson; and weak nuclear force, mediated by the $W^\pm$ and Z bosons. This work pertains to the description of the properties of the only family of particles that present no electric or color charge known as neutrinos (ν), wherein the SM exist three known flavors (ν_e , ν_μ and ν_τ) corresponding to the latter half of (electro)weak isospin doublets formed with the known charged leptons (e^- , μ^- and τ^-). This chapter aims to describe neutrinos in the SM, the weak force and its origin as a part of a larger broken symmetry known as "Electroweak Theory" (EW).

2.2. Symmetries and fermion content

2.2.1 Neutrinos

Neutrinos are charge-less, color-less, spin- $\frac{1}{2}$ particles with negligible masses compared to other fundamental particles first proposed in 1930 by Pauli W. [1], and later detected by C. L. Cowan et al. in 1956 [2], to solve known apparent contradictions in conserved quantities of beta decays. Beta decay was known to be as the emission of beta particles (electrons) by the nuclei of unstable atoms, wherein the difference in mass of the original and resulting nuclei of the decay should fix the energy at which the emitted particles are detected. Also of note, due to conservation of angular momentum the sum of the spin (S_e) of the electron, the spin of the resulting nucleus (S_{N^*}) and the angular momentum between them (L) should equal the original nucleus' spin (S_N). Despite this, it was known experimentally that the resulting electrons were emitted in a spectrum of energies less than expected and at

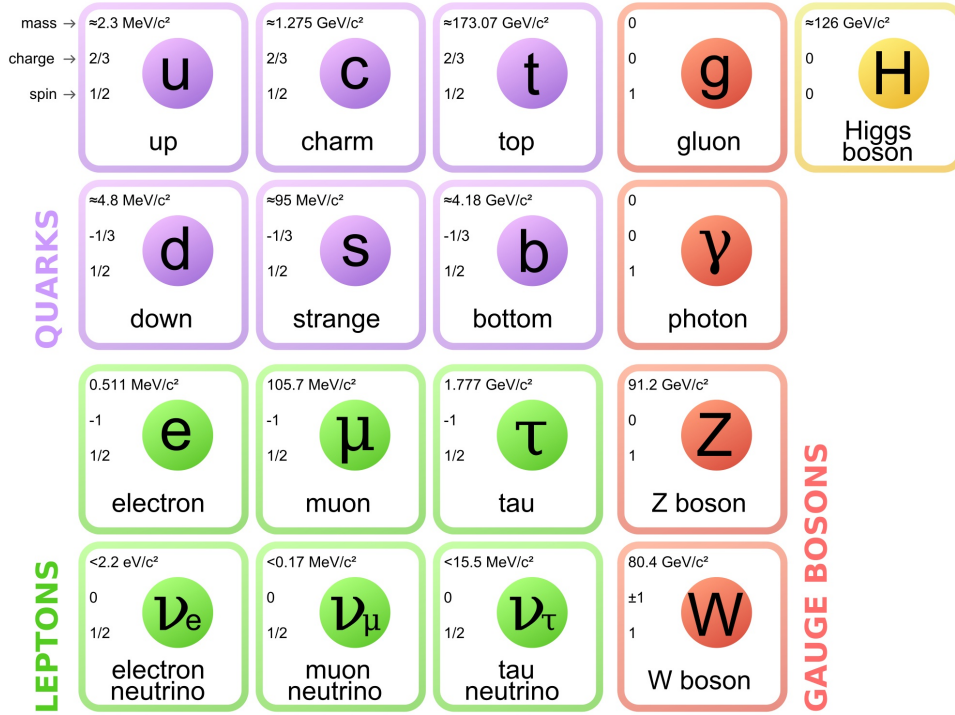


Figure 2.1: Particles in the SM

$L = 0$, making the sum of spins impossible to match. Pauli's solution to the apparent violation of the conservation of these quantities was to hypothesize the existence of a second neutral spin- $\frac{1}{2}$ with very small mass compared to the electron that allows for both the distribution of energy between both emitted particle, thus letting the electron be emitted at different energies depending on the kinematics, and the conservation of angular momentum.

$$\begin{array}{ccc}
 N \rightarrow N^* + e^- & \rightarrow & N \rightarrow N^* + e^- + \nu \\
 m_N = m_{N^*} + E_{e^-} & \rightarrow & m_N = m_{N^*} + E_{e^-} + E_\nu \\
 S_z N = S_z N^* + S_z e^- & \rightarrow & S_z N = S_z N^* + S_z e^- + S_z \nu
 \end{array}$$

Later refinement of the atomic model found that nuclei were made up of positively charged elements named protons (p) and neutrally charged elements known as neutrons¹ (n), leaving the small hypothesized particle to be named neutrino (ν) later by Enrico Fermi. It is well understood now that the beta decay is a process in which a neutron within an unstable nucleus decays into a proton, an electron and a(n) (anti)neutrino.

$$n \rightarrow p + e^- + \bar{\nu}$$

Subsequent experiments exploiting cross symmetry of the above interaction [2], accelerator-produced pion decay [3], and charged-current tau neutrino interactions in nuclear emulsion [4, 5] confirmed the existence of three distinct active neutrino flavors and the conservation of lepton family number in the interactions that involved these particles. Given that neutrinos don't interact via electromagnetic or strong forces a theory of a different force that mediated these interactions was developed into what is now known as the "weak nuclear force".

¹this was the name first proposed for the particle hypothesized by Pauli, later changed

2.2.2 Weak Nuclear Force

Developments in quantum mechanics (QM) and special relativity (SR) led up to the first theories that could describe the behavior of some subatomic particles and their interactions through excitations of quantized fields called quantum field theories (QFTs) obtained by promoting classical fields to operator-valued quantities satisfying canonical (anti)commutation relations and deriving their dynamics from a Lorentz-invariant action principle. One of the earliest and most successful QFTs was Quantum Electro Dynamics (QED), describing how spin-1/2 (anti)particles ($\bar{\psi}$) interacted with an EM 4-potential (A^μ) by minimizing the action given by the Lagrangian density:

$$\mathcal{L}_{\text{QED}} = \underbrace{\bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi}_{\text{free Dirac field}} - \underbrace{\frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{free EM field}} - \underbrace{e\bar{\psi}\gamma^\mu\psi A_\mu}_{\text{interaction}}. \quad (2.1)$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (2.2)$$

is the electromagnetic field strength tensor, ψ is the charged fermion field of mass m , and $D_\mu = \partial_\mu + ieA_\mu$ is the covariant derivative encoding the minimal coupling between the fermion and the gauge field A_μ . While the ‘free’ terms in (2.1) describe the propagation of Dirac fermions and photons in free Minkowski space, the ‘interaction’ term describes a vertex that couples the (anti)fermion current with the electromagnetic four-potential, representing the elementary process of a(n) (anti)fermion emitting or absorbing a photon. At zeroth-order $\hbar$ perturbative terms, QED had shown predictive power for processes like Compton ($\gamma + e^- \rightarrow \gamma + e^-$) [6] and Møller ($e^- + e^- \rightarrow e^- + e^-$) [7] scattering by the time Fermi first proposed an interaction inspired by the QED interaction term, where instead of a fermion current emits a photon, two fermion currents meet at a vertex such that a coupling constant (now called Fermi’s constant G_F) quantified the intensity of the interaction, and thus can be estimated experimentally.

$$\sim e \bar{\psi}_f \gamma^\mu \psi_f A_\mu$$

$$\sim \frac{G_F}{\sqrt{2}} (\bar{\psi}_n \gamma^\mu \psi_p) (\bar{\psi}_v \gamma_\mu \psi_{e^-})$$

Subsequent discoveries in particle physics revealed that charged-current weak interactions violate parity symmetry maximally. First theorized by Lee and Yang in 1956 [8] to resolve the then-known $\tau - \theta$ puzzle, and later verified experimentally by Wu et al. in 1957 [9], this motivated a modification of the Fermi interaction in the form of a combination of the two bilinear Lorentz-invariant structures that admit such violation [10, 11].

$$V : j_V^\mu = \bar{\psi} \gamma^\mu \psi \quad \xrightarrow{\mathcal{P}} \quad j'_{V\mu} = \bar{\psi} \gamma_\mu \psi \quad (2.3)$$

$$A : j_A^\mu = \bar{\psi} \gamma^\mu \gamma^5 \psi \quad \xrightarrow{\mathcal{P}} \quad j'_{A\mu} = -\bar{\psi} \gamma_\mu \gamma^5 \psi \quad (2.4)$$

The expressions (2.3) and (2.4) show how different bilinear structures change under parity ($\mathcal{P}$) and how pure vector (axial) currents are $\mathcal{P}$ -even ($\mathcal{P}$ -odd). Given a general current structure, $j^\mu = g_V j_V^\mu + g_A j_A^\mu$, observables that depend on the inner product of two such currents transform under parity as:

$$|\mathcal{M}|^2 = (j_1 \cdot j_2) = g_V^2 (j_{1V} \cdot j_{2V}) + g_A^2 (j_{1A} \cdot j_{2A}) + g_V g_A (j_{1V} \cdot j_{2A} + j_{1A} \cdot j_{2V}) \quad (2.5)$$

$$\xrightarrow{\mathcal{P}}$$

$$|\mathcal{M}'|^2 = (j'_1 \cdot j'_2) = g_V^2 (j'_{1V} \cdot j'_{2V}) + g_A^2 (j'_{1A} \cdot j'_{2A}) - g_V g_A (j'_{1V} \cdot j'_{2A} + j'_{1A} \cdot j'_{2V}). \quad (2.6)$$

The only difference between (2.5) and (2.6) lies in the sign of the interference term, such that the degree of $\mathcal{P}$ -symmetry violation is given by the coefficient $\frac{g_V g_A}{g_V^2 + g_A^2}$. Normalizing this there are two options for maximally violated $\mathcal{P}$ -symmetry given by $\frac{g_V}{g_A} = \pm 1$, also known as $V \pm A$ currents. Given that in the Wu experiment it was determined that (anti)neutrinos are emitted exclusively as left(right)-handed fermions the appropriate

structure to use for weak interactions was determined to be $V - A$. Modifying the Fermi Lagrangian density for beta decay to include this gives the following expression:

$$\mathcal{L}_{CC} = \frac{G_F}{\sqrt{2}} [\bar{\psi}_{\nu_e} \gamma^\mu (1 - \gamma^5) \psi_e] [\bar{\psi}_p \gamma_\mu (1 - \gamma^5) \psi_n] + \text{h.c.} \quad (2.7)$$

Later refinements of this theory succeeded in demonstrating the universality of the weak interaction across leptonic decays and strangeness-changing meson decays through Cabibbo mixing [12]. This not only established that all such processes are mediated by the same underlying force, but also motivated the prediction and subsequent discovery of the charm quark [13]. With these improvements on the theory thus far, the point-like structure of the interaction still led to scattering amplitudes proportional to the center-of-mass energy $\sqrt{s}$, resulting in S-matrix unitarity violation at higher energies. The natural resolution, in analogy with QED, is to conceive of Fermi's interaction as the low-energy limit of a gauge field theory, wherein the maximal parity violation of the charged currents naturally gives rise to three mediating vector bosons corresponding to the generators of an $SU(2)_L$ symmetry, which must be massive due to the short range of the weak interaction. Given this hypothesis, Glashow showed in 1961 [14] that the gauge field theory whose low-energy limit reproduces the $V - A$ structure of charged currents requires a $SU(2)_L \times U(1)_Y$ symmetry group, which not only predicted the existence of neutral weak currents, later measured in 1973 at the Gargamelle Neutrino Experiment at CERN [15, 16], but also unified the weak and electromagnetic interactions into a single electroweak theory, later completed independently by Weinberg and Salam through the incorporation of spontaneous symmetry breaking [17, 18].

2.3. Massive and Massless Mediators

2.3.1 Photons

It is well established within QED that electromagnetic interactions between charged fermions are mediated by a bosonic gauge field. The mediating boson, the photon, is massless as required by $U(1)$ gauge invariance, a property that manifests physically as the infinite range of the electromagnetic interaction, and carries spin-1. Its dynamics are governed by (2.1), and by applying the Euler-Lagrange equations to the free electromagnetic field term with respect to A^μ , one obtains the equations of motion:

$$\partial_\mu \left(\frac{\partial \mathcal{L}_{\text{free EM}}}{\partial (\partial_\mu A_\nu)} \right) - \frac{\partial \mathcal{L}_{\text{free EM}}}{\partial A_\nu} = 0 \implies \partial_\mu F^{\mu\nu} = \partial^2 A^\nu - \partial^\nu (\partial_\mu A^\mu) = 0, \quad (2.8)$$

which in momentum space, with $\partial_\mu \rightarrow -iq_\mu$, becomes

$$(-q^2 \eta^{\mu\nu} + q^\mu q^\nu) \tilde{A}_\nu = K^{\mu\nu} \tilde{A}_\nu = 0. \quad (2.9)$$

The propagator $D^{\mu\nu}$, defined by $K^{\mu\alpha} D_{\alpha\nu} = \delta^\mu_\nu$, cannot be obtained directly from the kinetic operator $K^{\mu\nu}$, since the latter is singular by construction: the action is invariant under $U(1)$ gauge transformations $A^\mu \rightarrow A'^\mu = A^\mu + \partial^\mu \lambda$, which implies that $K^{\mu\nu}$ has a zero eigenvalue along the gauge orbit. A propagator is only obtainable upon gauge

fixing, which is implemented by augmenting the Lagrangian density with a gauge-fixing term,

$$\mathcal{L}'_{\text{free EM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2, \quad (2.10)$$

such that the modified kinetic operator

$$K'^{\mu\nu} = -q^2\eta^{\mu\nu} + \left(1 - \frac{1}{\xi}\right)q^\mu q^\nu \quad (2.11)$$

is now invertible for any $\xi \neq 0$, yielding the generalized R_ξ propagator

$$D_\gamma^{\mu\nu}(q) = \frac{-i}{q^2 + i\epsilon} \left[\eta^{\mu\nu} - (1 - \xi) \frac{q^\mu q^\nu}{q^2} \right], \quad (2.12)$$

where any choice of $\xi \in \mathbb{R} \setminus \{0\}$ yields equivalent physical predictions, as gauge-dependent terms cancel in all observable quantities.

2.3.2 $W^\pm$ and Z

In contrast to the photon, the weak interaction's short range is a consequence of its mediators' mass and thus its dynamics are no longer governed by a Maxwell Lagrangian density, but rather one that includes a mass term that breaks the $U(1)$ found in the original, called a Proca Lagrangian, with a propagator that no longer requires gauge fixing but still is invariant under Lorentz transformations.

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2 A_\mu A^\mu. \quad (2.13)$$

In calculating its propagator by solving the equations of motion and inverting the kinetic operator in momentum space one finds that:

$$\partial_\mu \left(\frac{\partial \mathcal{L}_{\text{Proca}}}{\partial (\partial_\mu A_\nu)} \right) - \frac{\partial \mathcal{L}_{\text{Proca}}}{\partial A_\nu} = 0 \implies \partial_\mu F^{\mu\nu} + M^2 A^\nu = \partial^2 A^\nu - \partial^\nu (\partial_\mu A^\mu) + M^2 A^\nu = 0, \quad (2.14)$$

and by taking the four-divergence ∂_ν of (2.14),

$$\begin{aligned} \partial_\nu \partial^2 A^\nu - \partial_\nu \partial^\nu (\partial_\mu A^\mu) + M^2 (\partial_\mu A^\nu) &= 0, \\ \implies M^2 (\partial_\mu A^\nu) &= 0. \end{aligned} \quad (2.15)$$

Now taking $M^2 \neq 0$ the gauge given by $\partial_\mu A^\mu = 0$ is no longer a choice but rather an equation of motion that reduces the four degrees of freedom of the components of A^μ to two transverse and one longitudinal polarization modes of a massive spin-1 particle. Writing the equations of motion in momentum space with $\partial^\mu \rightarrow -iq^\mu$:

$$(-q^2\eta^{\mu\nu} + q^\nu q_\mu + M^2)A_\mu = ((M^2 - q^2)\eta^{\mu\nu} + q^\nu q_\mu)A_\mu = 0. \quad (2.16)$$

Inverting $K_{\text{Proca}}^{\mu\nu} = ((M^2 - q^2)\eta^{\mu\nu} + q^\mu q^\nu)$ gives a unique choice for the Proca propagator $D_{\text{Proca}}^{\mu\nu}$ such that $K_{\text{Proca}}^{\mu\alpha} D_{\text{Proca}}^{\alpha\nu} = \delta^\mu_\nu$.

$$D_{\text{Proca}}^{\mu\nu} = \frac{i}{q^2 - M^2 + i\epsilon} \left[\eta^{\mu\nu} - \frac{q^\mu q^\nu}{M^2} \right]. \quad (2.17)$$

It is worth noting that interactions mediated by (2.17) between conserved currents, for which $\partial_\mu j^\mu = 0$ implies $q_\mu \tilde{j}^\mu = 0$, reduce to the Fermi effective form in the limit $|q^2| \ll M^2$. Expanding the denominator,

$$\frac{1}{q^2 - M^2} = \frac{-1}{M^2} \left[\frac{1}{1 - q^2/M^2} \right] \xrightarrow{|q^2| \ll M^2} \frac{-1}{M^2} \left[1 + \frac{q^2}{M^2} + \mathcal{O}\left(\frac{q^4}{M^4}\right) \right], \quad (2.18)$$

where the amplitude of an interaction is now

$$i\mathcal{M} = (-ig)^2 j_\mu^\dagger D_{\text{Proca}}^{\mu\nu} j_\nu \xrightarrow{|q^2| \ll M^2} \frac{ig^2}{M^2} j_\mu^\dagger \eta^{\mu\nu} j_\nu, \quad (2.19)$$

which upon comparison with the Fermi four-fermion amplitude $i\mathcal{M}_{\text{Fermi}} = i\frac{G_F}{\sqrt{2}} j_\mu^\dagger \eta^{\mu\nu} j_\nu$ yields

$$\frac{G_F}{\sqrt{2}} \sim \frac{g^2}{M^2}, \quad (2.20)$$

where its actual matching relation will be given when calculating the $SU(2)_L$ couplings with $W^\pm$ and charged currents.

2.4. Electroweak mixing and $\sin^2 \theta_W$

2.4.1 $SU(2)_L$

As established in the previous section, the inclusion of massive spin-1 bosons explicitly breaks the usual $U(1)$ symmetry found in QED and thus a gauge theory that accommodates such mediators requires a larger symmetry group. A natural candidate comes in the study of a theory with an $SU(2)$ symmetry whose three generators,

$$\tau^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \tau^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \tau^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (2.21)$$

couple exclusively with left(right)-handed (anti)fermions (as is the case for charged current weak interactions) that act on doublets of the form

$$L = \begin{pmatrix} \nu_l \\ l \end{pmatrix}_L, \quad (2.22)$$

where l denotes any charged lepton and ν_l its corresponding (anti)neutrino. A theory of fermion interactions with a local $SU(2)_L$ symmetry requires promoting the partial derivative in the Dirac Lagrangian to a covariant derivative

$$D_\mu = \partial_\mu + ig \frac{\tau^a}{2} W_\mu^a = \partial_\mu + ig \left(\frac{\tau^1}{2} W_\mu^1 + \frac{\tau^2}{2} W_\mu^2 + \frac{\tau^3}{2} W_\mu^3 \right), \quad (2.23)$$

and introducing a non-abelian field strength tensor for each gauge boson W^a , $a \in \{1, 2, 3\}$,

$$W_{\mu\nu}^a = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a - g\epsilon^{abc} W_\mu^b W_\nu^c, \quad (2.24)$$

where the last term, absent in the abelian case of QED, encodes the self-interactions among the gauge bosons that arise from the non-abelian nature of $SU(2)_L$. The resulting Lagrangian density is

$$\mathcal{L}_{SU(2)_L} = \bar{L} i \gamma^\mu D_\mu L - \frac{1}{4} W_{\mu\nu}^a W^{a\mu\nu}, \quad (2.25)$$

where $L = \begin{pmatrix} \nu_l \\ l \end{pmatrix}_L$ denotes the left-handed lepton doublet. Extracting and expanding the interaction term in (2.25) explicitly gives,

$$\begin{aligned} \mathcal{L}_{SU(2)_L, \text{int}} &= -\frac{g}{2} \bar{L} \gamma^\mu (\tau^1 W_\mu^1 + \tau^2 W_\mu^2 + \tau^3 W_\mu^3) L \\ &= -\frac{g}{2} (\bar{\nu}_l \quad \bar{l})_L \gamma^\mu \begin{pmatrix} W_\mu^3 & W_\mu^1 - i W_\mu^2 \\ W_\mu^1 + i W_\mu^2 & -W_\mu^3 \end{pmatrix} \begin{pmatrix} \nu_l \\ l \end{pmatrix}_L. \end{aligned} \quad (2.26)$$

Identifying that the upper right and lower left entries of the matrix in (2.26) correspond neatly to the known charged current weak interactions makes the definition of the $W^\pm$ bosons such that $W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp i W_\mu^2)$ a way to separate the interaction Lagrangian density into a charged current (CC) term and a neutral current (NC) term.

$$\mathcal{L}_{\text{int}} = \underbrace{-\frac{g}{2} [\bar{\nu}_L \gamma^\mu \nu_L - \bar{l}_L \gamma^\mu l_L] W_\mu^3}_{\mathcal{L}_{\text{NCint}}} - \underbrace{\frac{g}{\sqrt{2}} [\bar{\nu}_L \gamma^\mu l_L W_\mu^+ + \bar{l}_L \gamma^\mu \nu_L W_\mu^-]}_{\mathcal{L}_{\text{CCint}}}. \quad (2.27)$$

It is worth noting that the amplitude of a CC interaction computed from $\mathcal{L}_{\text{CCint}}$ in (2.27) in the limit $|q^2| \ll M_W^2$ reproduces the Fermi four-fermion amplitude, yielding the tree-level matching relation between G_F and g .

$$-i\mathcal{M} = \left[\frac{-ig}{\sqrt{2}} \bar{\nu}_L \gamma^\mu l_L \right] \frac{i\eta^{\mu\nu}}{M_W^2 - q^2} \left[\frac{-ig}{\sqrt{2}} \bar{l}_L \gamma_\nu \nu_L \right] \quad (2.28)$$

$$\begin{aligned} &= \left[\frac{-ig}{\sqrt{2}} \bar{\nu} \frac{1}{2} \gamma_\mu (1 - \gamma^5) l \right] \frac{i\eta^{\mu\nu}}{M_W^2 - q^2} \left[\frac{-ig}{\sqrt{2}} \bar{l} \gamma_\nu (1 - \gamma^5) \nu \right] \\ &\stackrel{|q^2| \ll M_W^2}{\approx} -\frac{ig^2}{8M_W^2} [\bar{\nu} \gamma^\mu (1 - \gamma^5) l] [\bar{l} \gamma_\mu (1 - \gamma^5) \nu]. \end{aligned} \quad (2.29)$$

Comparing (2.29) with the amplitude given by (2.7) yields the relation

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}, \quad (2.30)$$

which is of particular importance as it expresses the experimentally well-measured Fermi constant directly in terms of the $SU(2)_L$ gauge coupling and the $W^\pm$ mass, thereby connecting the low-energy effective theory to the underlying electroweak gauge structure and providing a precise prediction for M_W . Given this and the knowledge that weak neutral currents exist, it is tempting to identify W_μ^3 as their mediator. However, this cannot be the case since this term couples exclusively to left(right)-handed (anti)fermions, while it is experimentally well established that weak neutral current interactions do not maximally violate $\mathcal{P}$ -symmetry as charged current interactions do [15, 16]. The $SU(2)_L$ symmetry is therefore too chiral to

Multiplet	T^3	Q	Y
ν_{lL}	$\frac{1}{2}$	0	-1
l_L	$-\frac{1}{2}$	-1	-1
l_R	0	-1	-2

Table 2.1: Values for Q , T^3 and Y for each lepton, where l denotes any of the known flavours of charged lepton

accommodate weak neutral currents, which while not $\mathcal{P}$ -conserving, do not exhibit the maximal parity violation of the charged current sector. This incompatibility however points to the possibility of a more complete theory wherein the purely chiral $SU(2)_L$ generators are combined with a $\mathcal{P}$ -conserving $U(1)_Y$ hypercharge gauge field, analogous to that of QED, to model the weak and electromagnetic interactions as a single unified force known as the Electroweak (EW) theory.

2.4.2 $SU(2)_L \times U(1)_Y$

To establish a theory coherent with the known properties of leptons, it is first necessary to fix the hypercharge (Y) assignments of the left-handed fermion doublet $L = \begin{pmatrix} \nu_l \\ l \end{pmatrix}_L$ and the right-handed lepton singlet $R = (l_R)$ by relating them to the known electric charges Q and weak isospin eigenvalues T^3 via the Gell-Mann–Nishijima relation

$$Q = T^3 + \frac{Y}{2}. \quad (2.31)$$

Fixing these gives the values presented in Table 2.1 for each lepton type, noting that right-handed neutrinos are absent from the spectrum due to the lack of experimental evidence for their existence, with the exception of the measured neutrino mass splittings among the three flavors, which falls outside the scope of this work. With this, the full Lagrangian density with the necessary assigned values is:

$$\mathcal{L}_{EW} = -i\bar{L}\gamma^\mu D_\mu L - i\bar{R}\gamma^\mu D_\mu R - \frac{1}{4}W^{a\mu\nu}W_{\mu\nu}^a - \frac{1}{4}B^{\mu\nu}B_{\mu\nu} \quad (2.32)$$

Where $B^{\mu\nu}$ is defined the same way as $F^{\mu\nu}$ in (2.2) for a massless boson field B^μ . The covariant derivative is given by

$$D_\mu = \partial_\mu + ig\left(\frac{\tau^1}{2}W_\mu^1 + \frac{\tau^2}{2}W_\mu^2 + \frac{\tau^3}{2}W_\mu^3\right) + ig'\frac{Y}{2}B_\mu. \quad (2.33)$$

Much like (2.26), the interaction terms of (2.32) can be separated into CC and NC contributions, wherein the NC terms can be re-expressed as a linear combination of two mass eigenstates mixed through an orthogonal rotation parametrized by a single angle known as the Weinberg angle (θ_W). The EW interaction terms are given by

$$\begin{aligned} \mathcal{L}_{EW \text{ int}} &= -\frac{g}{2} (\bar{\nu}_l \quad \bar{l})_L \gamma^\mu \begin{pmatrix} W_\mu^3 & W_\mu^1 - iW_\mu^2 \\ W_\mu^1 + iW_\mu^2 & -W_\mu^3 \end{pmatrix} \begin{pmatrix} \nu_l \\ l \end{pmatrix}_L - \frac{g'}{2} (\bar{\psi}\gamma^\mu Y \psi) B_\mu \\ &= \underbrace{-\frac{g}{\sqrt{2}} \left[\bar{\nu}_L \gamma^\mu l_L W_\mu^+ + \bar{l}_L \gamma^\mu \nu_L W_\mu^- \right]}_{\mathcal{L}_{CC \text{ int}}} \underbrace{-\frac{g}{2} [\bar{\nu}_L \gamma^\mu \nu_L - \bar{l}_L \gamma^\mu l_L] W_\mu^3 - \frac{g'}{2} \bar{\psi} \gamma^\mu Y B_\mu \psi}_{\mathcal{L}_{NC \text{ int}}}, \end{aligned} \quad (2.34)$$

where ψ denotes fermions with hypercharge Y and weak isospin T^3 , the latter being explicitly dependent on the chirality of the field, as left-handed fermions carry $T^3 = \pm 1/2$ while right-handed fermions are $SU(2)_L$ singlets with $T^3 = 0$.

$$\begin{aligned}\mathcal{L}_{NC \text{ int}} &= -\frac{g}{2} [\bar{\nu}_L \gamma^\mu \nu_L - \bar{l}_L \gamma^\mu l_L] W_\mu^3 - \frac{g'}{2} \bar{\psi} \gamma^\mu Y B_\mu \psi \\ &= -\frac{g}{2} \bar{\psi} \gamma^\mu (2T^3) \psi - \frac{g'}{2} \bar{\psi} \gamma^\mu Y B_\mu \psi \\ &= -\bar{\psi} \gamma^\mu (g T^3 W_\mu^3 + g' \frac{Y}{2} B_\mu) \psi.\end{aligned}\tag{2.35}$$

The W_μ^3 and B_μ fields correspond to the unmixed states that can be combined into the known mass eigenstates A_μ and Z_μ through the orthogonal transformation given by:

$$\begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix} = \begin{pmatrix} \cos \theta_W & -\sin \theta_W \\ \sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix}.\tag{2.36}$$

Substituting (2.36) into (2.35),

$$\begin{aligned}\mathcal{L}_{NC \text{ int}} &= -\left(g \sin \theta_W T^3 + g' \cos \theta_W \frac{Y}{2} \right) \bar{\psi} \gamma^\mu \psi A_\mu \\ &\quad - \left(g \cos \theta_W T^3 - g' \sin \theta_W \frac{Y}{2} \right) \bar{\psi} \gamma^\mu \psi Z_\mu.\end{aligned}\tag{2.37}$$

Requiring that A_μ corresponds to the photon field that couples with fermions such that $\mathcal{L}_{QED \text{ int}} = -e \bar{\psi} \gamma^\mu Q \psi$ where Q follows from (2.31) gives the relations between the coupling constants

$$e = g \sin \theta_W = g' \cos \theta_W.\tag{2.38}$$

Substituting (2.31) and (2.38) into the Z_μ coefficient of (2.37) yields the Z boson interaction term expressed solely in terms of T^3 and Q ,

$$\mathcal{L}_{NC \text{ int}} = -e Q \bar{\psi} \gamma^\mu \psi A_\mu - \frac{g}{\cos \theta_W} (T^3 - Q \sin^2 \theta_W) \bar{\psi} \gamma^\mu \psi Z_\mu.\tag{2.39}$$

It will be useful to later sections of this work to have the Z boson interaction term separated into vector and axial-vector terms such that for a fermion f ,

$$\mathcal{L}_{Z f \bar{f} \text{ int}} = -\frac{g}{\cos \theta} \bar{\psi}_f \gamma^\mu (g_V^f - g_A^f \gamma^5) \psi_f Z_\mu.\tag{2.40}$$

The vector (g_V^f) and axial-vector (g_A^f) coupling coefficients can be calculated explicitly noting that all right-handed fermions are $SU(2)_L$ singlets and thus $T^3 P_R \psi = T^3 \frac{1}{2} (1 + \gamma^5) \psi = T^3 \psi_R = 0$. Separating the Z interaction term in (2.39),

$$\begin{aligned}\mathcal{L}_{Z f \bar{f} \text{ int}} &= -\frac{g}{\cos \theta_W} \left(T_f^3 - Q_f \sin^2 \theta_W \right) \bar{\psi}_f \gamma^\mu \psi_f Z_\mu \\ &= -\frac{g}{\cos \theta_W} \bar{\psi}_f \frac{\gamma^\mu}{2} \left(T_f^3 - Q_f \sin^2 \theta_W \right) (1 + 1 + \gamma^5 - \gamma^5) \psi_f Z_\mu \\ &= -\frac{g}{\cos \theta_W} \bar{\psi}_f \frac{\gamma^\mu}{2} \left(\cancel{T_f^3}^0 - Q_f \sin^2 \theta_W \right) (1 + \gamma^5) \psi_f Z_\mu\end{aligned}$$

$$\begin{aligned}
& -\frac{g}{\cos \theta_W} \bar{\psi}_f \frac{\gamma^\mu}{2} \left(T_f^3 - Q_f \sin^2 \theta_W \right) (1 - \gamma^5) \psi_f Z_\mu \\
& = -\frac{g}{\cos \theta_W} \bar{\psi}_f \gamma^\mu \left(\left(\frac{T_f^3}{2} - Q_f \sin^2 \theta_W \right) - \frac{T_f^3}{2} \gamma^5 \right) \psi_f Z_\mu.
\end{aligned} \tag{2.41}$$

Comparing (2.41) with (2.40),

$$\left\{ \begin{aligned} g_V^f &= \frac{T_f^3}{2} - Q_f \sin^2 \theta_W, \end{aligned} \right. \tag{2.42}$$

$$\left\{ \begin{aligned} g_A^f &= \frac{T_f^3}{2}. \end{aligned} \right. \tag{2.43}$$

It is worth noting that the presence of the $Q_f \sin^2 \theta_W$ term in g_V^f (??) is precisely what prevents the Z boson interaction from being maximally $\mathcal{P}$ -violating. The couplings g_V^f and g_A^f constitute the tree-level SM predictions for the vector and axial-vector couplings of neutral current weak interactions, whose values are fixed entirely by T^3 , Q , and $\sin^2 \theta_W$ for each fermion. However, current experimental precision in measuring observables of such interactions, most notably in neutrino experiments, has not only confirmed these tree-level predictions but also demands that higher-order perturbative corrections in the form of loop diagrams be taken into account. These radiative corrections introduce a dependence on the momentum transfer Q^2 , effectively promoting $\sin^2 \theta_W$ to a running coupling constant as will be discussed in Section 3.5 whose value differs measurably between the Z pole and the low-energy regime probed by reactor neutrino experiments, and acquire physical meaning when interpreted as corrections to the $ff\gamma$ vertex function. The following chapter focuses on how, despite carrying no electric charge, the neutrino nonetheless possesses measurable electromagnetic properties predicted by the SM through precisely these radiative corrections.

3. Radiative Corrections and Neutrino Charge Radius

This chapter is concerned with the main theoretical focus of this work. The SM as described in Chapter 2, and in specific EW theory within it, models the interactions between fermions and gauge bosons utilizing of perturbative terms in the form of higher-order loop diagrams. While tree level calculations offered good agreement with experimental data, modern precision in particle experiments have required in recent history have shown the necessity to include higher-order terms to either validate the SM or in occasions point to scenarios Beyond the SM (BSM). This chapter describes how the inclusion of 1-loop diagrams to an effective interaction reveal predictions of electromagnetic properties of the neutrino and how they modify previously established parameters in the SM.

3.1. Introduction

The electromagnetic properties of the neutrino were not only essential to its first postulation and subsequent discovery, but remain an active field of research in modern experimental particle physics. Indeed, both its neutral charge and the possibility of a non-zero magnetic dipole moment are mentioned in the original 1930 letter by Pauli that first introduced the neutrino [1]. The study of such properties was further motivated by the proposal and eventual discovery of neutrino flavor oscillations and their requisite non-zero mass splittings. This chapter describes how radiative corrections to the effective $\nu\nu\gamma$ vertex predict non-trivial electromagnetic form factors for massive¹ neutrinos within the SM, the focus of this work being the neutrino charge radius (NCR).

3.2. Electromagnetic Form Factors

3.2.1 Massive Neutrinos

Following Ref. [21], the interaction between a fermionic field $\bar{f}(x)$ and the electromagnetic field A^μ in the SM is given by the interaction Hamiltonian

$$\mathcal{H}_{\text{EM}}(x) = j_\mu^\dagger(x)A^\mu(x) = q_f \bar{f}(x)\gamma_\mu f(x)A^\mu(x), \quad (3.1)$$

where q_f is the electric charge of a fermion f . Fig. 3.1(a) shows the corresponding the tree-level Feynman diagram for (3.1) where the photon γ is the quanta of the EM field A^μ . For the neutrino no such direct interaction term exists due to its neutral charge. Nevertheless, at the quantum level such interactions can arise when including higher order terms in the perturbative expansion of interaction terms in the form of loop

¹This chapter concerns primarily Dirac-type neutrinos; analogous calculations for Majorana neutrinos can be found in [19, 20, 21]

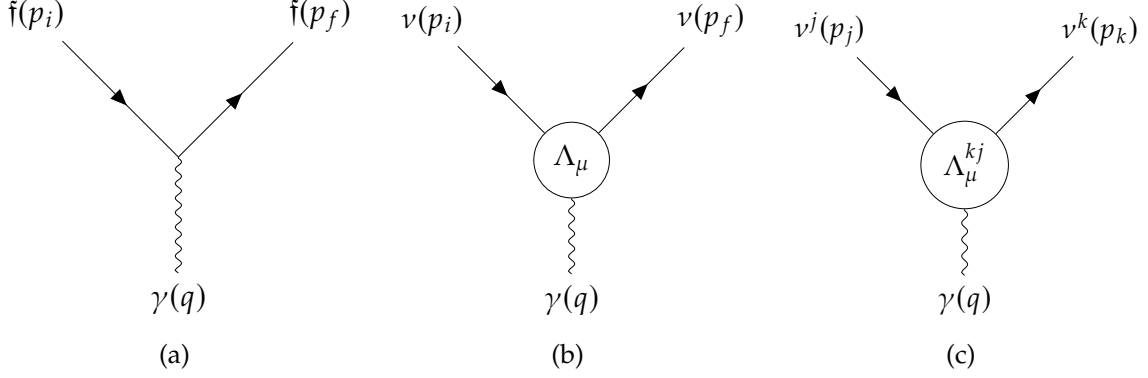


Figure 3.1: my caption

diagrams. In the 1-photon approximation the effective interaction Hamiltonian for a massive neutrino ν is given by

$$\mathcal{H}_{\text{EM}}^{(\nu)}(x) = j_\mu^{(\nu)}(x)A^\mu(x) = \bar{\nu}(x)\Lambda_\mu\nu(x)A^\mu(x), \quad (3.2)$$

where $j_\mu^{(\nu)}(x)$ is the effective neutrino current and Λ_μ is a 4×4 vertex function matrix in spinor space which can contain space-time derivatives such that $j_\mu^{(\nu)}(x)$ transforms as a four-vector. The corresponding effective Feynman diagram for (3.2) is shown in Fig. 3.1(b). We will see that the neutrino electromagnetic form factors and properties such as charge and magnetic dipole will correspond to the Fig. 3.1(b) interaction, but it is worth contrasting the origin of these form factors with the analogous quantities for composite particles such as the proton or the neutron, where electromagnetic form factors encode genuine information about the spatial distribution of fundamental constituents. The neutrino carries no such substructure; its form factors are instead radiative effects generated by quantum loops involving gauge bosons. They are physical and, in principle, measurable, but they constitute effective electromagnetic properties induced by the weak interaction rather than a reflection of any internal composition. In this sense the neutrino electromagnetic form factors occupy a conceptually distinct place within the Standard Model.

We are interested in the neutrino part of the Fig. 3.1(b) interaction amplitude, given by the matrix element

$$\langle \nu(p_f, h_f) | j_\mu^{(\nu)}(x) | \nu(p_i, h_i) \rangle, \quad (3.3)$$

where $p_{i(f)}$ and $h_{i(f)}$ denote the momentum and helicity of the initial (final) neutrino respectively. Setting aside the helicity dependence for now and taking into account

$$\partial^\mu j_\mu^{(\nu)} = i \left[\mathcal{P}^\mu, j_\mu^{(\nu)} \right], \quad (3.4)$$

where $\mathcal{P}^\mu$ is the four-momentum operator which generates translations, the effective neutrino current can be rewritten as

$$j_\mu^{(\nu)}(x) = e^{i\mathcal{P} \cdot x} j_\mu^{(\nu)}(0) e^{-i\mathcal{P} \cdot x}. \quad (3.5)$$

Since $\mathcal{P}^\mu | \nu(p_{i(f)}) \rangle = p_{i(f)}^\mu | \nu(p_{i(f)}) \rangle$ substituting (3.5) into (3.3) gives

$$\langle \nu(p_f) | j_\mu^{(\nu)}(x) | \nu(p_i) \rangle = e^{(p_f - p_i) \cdot x} \langle \nu(p_f) | j_\mu^{(\nu)}(0) | \nu(p_i) \rangle. \quad (3.6)$$

It is evident from (3.6) that the quantity encoding the neutrino–photon interaction is the matrix element $\langle \nu(p_f) | j_\mu^{(\nu)}(0) | \nu(p_i) \rangle$. Since $|\nu(p_i)\rangle$ and $\langle \nu(p_f)|$ describe incoming and outgoing free Dirac particles whose field admits the Fourier expansion

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3 2E} \sum_{h=\pm} [a^{(h)}(p) u^{(h)}(p) e^{-ix \cdot p} + b^{(h)\dagger}(p) v^{(h)}(p) e^{ix \cdot p}], \quad (3.7)$$

the only spinor structures that survive in (3.6) are those sandwiched between free-particle spinors, so that

$$\langle \nu(p_f) | j_\mu^{(\nu)}(0) | \nu(p_i) \rangle = \bar{u}(p_f) \Lambda_\mu(p_f, p_i) u(p_i). \quad (3.8)$$

Here Λ_μ is a 4×4 matrix in spinor space whose free Lorentz index must transform as a contravariant four-vector. The most general such matrix constructed from γ -matrices and the available four-momenta $p_{i,f}$, or equivalently from $P_\mu = (p_f + p_i)_\mu$ and $q_\mu = (p_f - p_i)_\mu$, is [22]

$$\Lambda_\mu = \mathbb{f}_1(q^2) q_\mu + \mathbb{f}_2(q^2) q_\mu \gamma_5 + \mathbb{f}_3(q^2) \gamma_\mu + \mathbb{f}_4(q^2) \gamma_\mu \gamma_5 + \mathbb{f}_5(q^2) \sigma_{\mu\nu} q^\nu + \mathbb{f}_6(q^2) \epsilon_{\mu\nu\rho\lambda} q^\nu \sigma^{\rho\lambda}, \quad (3.9)$$

where $\mathbb{f}_k$ ($k = 1, \dots, 6$) are six Lorentz-invariant form factors and q is the four-momentum transferred to the photon. Its worth noting that q^2 is the only available independent Lorentz-invariant kinematic quantity since $(p_i + p_f)^2 = 4m_\nu^2 - q^2$ and thus the vertex function $\Lambda_\mu(q)$ can be written as only dependent on q . Further restrictions on $\Lambda_\mu(q)$ are seen when taking into account that $\mathcal{H}_{\text{EM}}^{(\nu)} = \mathcal{H}_{\text{EM}}^{(\nu)\dagger}$ and $A_\mu = A_\mu^\dagger$ thus

$$\langle \nu(p_f) | j_\mu^{(\nu)}(0) | \nu(p_i) \rangle = \langle \nu(p_f) | j_\mu^{(\nu)}(0) | \nu(p_i) \rangle^\dagger \quad (3.10)$$

$$\implies \bar{u}(p_f) \Lambda_\mu(q) u(p_i) = (\bar{u}(p_f) \Lambda_\mu(q) u(p_i))^\dagger \quad (3.11)$$

$$\implies \Lambda_\mu(q) = \gamma^0 \Lambda_\mu^\dagger(-q) \gamma^0. \quad (3.12)$$

Substituting (3.12) into (3.9) constraints the form factor functions such that

$$\mathbb{f}_2, \mathbb{f}_3, \mathbb{f}_4 \text{ are real} \quad (3.13)$$

and

$$\mathbb{f}_1, \mathbb{f}_5, \mathbb{f}_6 \text{ are imaginary.} \quad (3.14)$$

Furthermore, imposing current conservation $\partial^\mu j_\mu^{(\nu)} = 0$ required by the EM $U(1)$ gauge invariance on $A^\mu \rightarrow A^\mu + \partial^\mu \varphi(x)$ for any $\varphi(x)$ which leaves (2.2) invariant in combination with (3.4) leads to

$$\langle \nu(p_f) | [\mathcal{P}^\mu, j_\mu^{(\nu)}(0)] | \nu(p_i) \rangle = 0 \quad (3.15)$$

$$\implies q^\mu \bar{u}(p_f) \Lambda_\mu(q) u(p_i) = 0. \quad (3.16)$$

Utilizing on-shell conditions for the neutrino spinors $\not{p}_{i(f)} u(p_{i(f)}) = m u(p_{i(f)})$, the antisymmetry of $\sigma_{\mu\nu}$ and $\epsilon_{\mu\nu\rho\gamma}$ and the anticommutation relation of γ_μ with γ_5

$$q^\mu \bar{u}(p_f) \Lambda_\mu(q) u(p_i) = \bar{u}(p_f) [\mathbb{f}_1(q^2) q^\mu q_\mu + \mathbb{f}_2(q^2) q^\mu q_\mu \gamma_5 + \mathbb{f}_3(q^2) \overbrace{q^\mu \gamma_\mu}^{q = \not{p}_i - \not{p}_f}]$$

$$\begin{aligned}
& + \mathbb{f}_4(q^2) \underbrace{q^\mu \gamma_\mu}_{q = \not{p}_i - \not{p}_f} \gamma_5 + \mathbb{f}_5(q^2) \cancel{q^\mu \sigma_{\mu\nu} q^\nu} + \mathbb{f}_6(q^2) \cancel{q^\mu \epsilon_{\mu\nu\rho\lambda} q^\nu \sigma^{\rho\lambda}}] u(p_i) \\
& = \bar{u}(p_f) [\mathbb{f}_1(q^2) q^2 + \mathbb{f}_2(q^2) q^2 \gamma_5 \\
& + \mathbb{f}_3(q^2) (\not{p}_i - \not{p}_f) + \mathbb{f}_4(q^2) (\not{p}_i - \not{p}_f) \gamma_5] u(p_i) \\
& = \bar{u}(p_f) [\mathbb{f}_1(q^2) q^2 + \mathbb{f}_2(q^2) q^2 \gamma_5 \\
& + \mathbb{f}_3(q^2) \cancel{(m - m)} + \mathbb{f}_4(q^2) (m + m) \gamma_5] u(p_i) \\
& = \bar{u}(p_f) [\mathbb{f}_1(q^2) q^2 + \mathbb{f}_2(q^2) q^2 \gamma_5 + 2m \mathbb{f}_4(q^2) \gamma_5] u(p_i). \tag{3.17}
\end{aligned}$$

Substituting (3.17) into (3.16)

$$\implies \mathbb{f}_1(q^2) q^2 + \mathbb{f}_2(q^2) q^2 \gamma_5 + 2m \mathbb{f}_4(q^2) \gamma_5 = 0. \tag{3.18}$$

Taking into account that $\mathbb{I}$ and γ_5 are linearly independent leaves $\mathbb{f}_1(q^2) = 0$ and $\mathbb{f}_4(q^2) = -\frac{q^2}{2m} \mathbb{f}_2(q^2)$. Substituting back into (3.9) leaves

$$\Lambda_\mu(q) = \mathbb{f}_Q(q^2) \gamma_\mu - \mathbb{f}_M(q^2) i \sigma_{\mu\nu} q^\nu + \mathbb{f}_E(q^2) \sigma_{\mu\nu} q^\nu \gamma_5 + \mathbb{f}_A(q^2) (q^2 \gamma_\mu - q_\mu \not{q}) \gamma_5, \tag{3.19}$$

where $\mathbb{f}_Q = \mathbb{f}_3$, $\mathbb{f}_M = i \mathbb{f}_5$, $\mathbb{f}_E = -2i \mathbb{f}_6$ and $\mathbb{f}_A = -\mathbb{f}_2/2m$ are the real charge, dipole magnetic and electric, and anapole neutrino form factors. For the coupling with the real photon at $q^2 = 0$

$$\mathbb{f}_Q(0) = \mathfrak{q}, \quad \mathbb{f}_M(0) = \mu, \quad \mathbb{f}_E(0) = \epsilon, \quad \mathbb{f}_A(0) = \mathfrak{a}, \tag{3.20}$$

where $\mathfrak{q}$ is the electric charge, μ the magnetic dipole, ϵ the electric dipole and $\mathfrak{a}$ the anapole moment of the neutrino.

3.2.2 Massive Neutrino Mixing

It is worth noting that we have only considered the effective interaction of a single mass eigenstate neutrino with the photon, whereas a full treatment of the known mass mixing among neutrinos requires promoting the aforementioned vertex function and form factors into $N \times N$ matrices, where $\{|v_k\rangle\}$, $k = 1, 2, \dots, N$, describe N neutrino mass eigenstates of definite mass satisfying $(i\not{\partial} - m_k)v_k = 0$. A generalized version of (3.2), described by the effective Feynman diagram in Fig. 3.1(c), is given by

$$\mathcal{H}_{\text{EM}}^{(\nu)}(x) = j_\mu^{(\nu)}(x) A^\mu(x) = \sum_{k,j=1}^N \bar{v}_k(x) \Lambda_\mu^{kj} v_j(x) A^\mu(x), \tag{3.21}$$

where, for an initial state $|v_i(p_i)\rangle$ and a final state $|v_f(p_f)\rangle$ describing free mass-eigenstate neutrinos, the generalized form of the matrix element in (3.8) reads

$$\langle v_f(p_f) | j_\mu^{(\nu)}(0) | v_i(p_i) \rangle = \bar{v}_f(p_f) \Lambda_\mu^{fi}(p_f, p_i) v_i(p_i). \tag{3.22}$$

As was the case for $\Lambda_\mu(p_f, p_i)$, the matrix $\Lambda_\mu^{fi}(p_f, p_i)$ depends only on q for the same underlying reason, and in its most general form is a linear combination of the same covariant elements, namely

$$\begin{aligned}\Lambda_\mu^{fi} = & \mathbb{f}_1^{fi}(q^2) q_\mu + \mathbb{f}_2^{fi}(q^2) q_\mu \gamma_5 + \mathbb{f}_3^{fi}(q^2) \gamma_\mu \\ & + \mathbb{f}_4^{fi}(q^2) \gamma_\mu \gamma_5 + \mathbb{f}_5^{fi}(q^2) \sigma_{\mu\nu} q^\nu + \mathbb{f}_6^{fi}(q^2) \epsilon_{\mu\nu\rho\lambda} q^\nu \sigma^{\rho\lambda},\end{aligned}\quad (3.23)$$

with conditions (3.13) and (3.14) holding for the corresponding $\mathbb{f}_m^{fi}$, $m = 1, \dots, 6$. Imposing current conservation, the on-shell conditions for the neutrino spinors $\bar{v}_f \not{q} v_i = \bar{v}_f (m_f - m_i) v_i$ and the algebraic properties of the covariant forms appearing in (3.23) yields the constraints

$$\begin{cases} \mathbb{f}_1^{fi}(q^2) + (m_f - m_i) \mathbb{f}_3^{fi}(q^2) = 0, \\ \mathbb{f}_2^{fi}(q^2) + (m_f + m_i) \mathbb{f}_4^{fi}(q^2) = 0. \end{cases}\quad (3.24)$$

$$\quad (3.25)$$

so that the vertex function (3.19) is correspondingly promoted to

$$\Lambda_\mu^{fi}(q) = (\gamma_\mu - q_\mu \not{q} / q^2) (\mathbb{f}_Q^{fi}(q^2) + \mathbb{f}_A^{fi}(q^2) q^2 \gamma_5) - i \sigma_{\mu\nu} q^\nu (\mathbb{f}_M^{fi}(q^2) + i \mathbb{f}_E^{fi}(q^2) \gamma_5), \quad (3.26)$$

where $\mathbb{f}_Q^{fi} = \mathbb{f}_3^{fi}$, $\mathbb{f}_M^{fi} = i \mathbb{f}_5^{fi}$, $\mathbb{f}_E^{fi} = -2i \mathbb{f}_6^{fi}$, and $\mathbb{f}_A^{fi} = -\mathbb{f}_2^{fi} / (m_f + m_i)$, subject to the Hermiticity condition

$$\mathbb{f}_\Omega^{fi} = (\mathbb{f}_\Omega^{if})^* \quad (\Omega = Q, E, M, A). \quad (3.27)$$

Coupling with the real photon at $q^2 = 0$

$$\mathbb{f}_Q^{fi}(0) = \mathbf{q}^{fi}, \quad \mathbb{f}_M^{fi}(0) = \mu^{fi}, \quad \mathbb{f}_E^{fi}(0) = \epsilon^{fi}, \quad \mathbb{f}_A^{fi}(0) = \mathbf{a}^{fi}, \quad (3.28)$$

where $\mathbf{q}^{fi}$ is the electric charge, μ^{fi} the magnetic dipole, ϵ^{fi} the electric dipole and $\mathbf{a}^{fi}$ the anapole transition moment of an initial $|v_i\rangle$ and final $|v_f\rangle$ neutrino mass eigenstate.

It is worth noting that the diagonal entries $f = i$ of $\mathbb{f}_\Omega^{fi}$ and $\Lambda_\mu^{fi}(q)$ reduce to their single mass-eigenstate counterparts found previously, whereas the off-diagonal entries $f \neq i$ describe transition electromagnetic properties between distinct mass eigenstates, a feature entirely absent in the single-flavor treatment and one of the principal new phenomena introduced by neutrino mixing.

3.2.3 Flavor Base

It is important to recognize, however, that the mass eigenstates $|v_i\rangle$ are not the states that are produced or detected in weak interaction processes; rather, it is the flavor eigenstates $|v_\alpha\rangle$, $\alpha = e, \mu, \tau$, that couple directly to the charged leptons through the $W^\pm$ boson, the two bases being related by the leptonic mixing matrix U through $|v_\alpha\rangle = \sum_i U_{\alpha i}^* |v_i\rangle$, better known as the PMNS matrix [23, 24]. Consequently, the

electromagnetic vertex function in the flavor basis, $\tilde{\Lambda}_\mu^{\alpha\beta}$, is obtained from Λ_μ^{ij} by the corresponding unitary rotation

$$\tilde{\Lambda}_\mu^{\alpha\beta}(q) = \sum_{i,j=1}^3 U_{\alpha i} \Lambda_\mu^{ij}(q) U_{\beta j}^*, \quad (3.29)$$

with the inverse relation $\Lambda_\mu^{ij} = \sum_{\alpha,\beta} U_{\alpha i}^* \tilde{\Lambda}_\mu^{\alpha\beta} U_{\beta j}$ following immediately from the unitarity of U . While the algebraic structure of (3.26) is preserved under this rotation and thus the EM form factors can be defined in the flavor basis such that

$$\tilde{f}_\Omega^{\alpha\beta}(q^2) = \sum_{i,j=1}^3 U_{\alpha i} \mathbb{f}_\Omega^{ij}(q^2) U_{\beta j}^*, \quad (3.30)$$

the flavor basis offers no particular simplification on its own; rather, its utility becomes apparent once the calculation of the SM electroweak loop diagrams generating Λ_μ is undertaken, since it is precisely in this basis that the relevant interaction vertices are diagonal.

3.2.4 Weyl Neutrinos

A further simplification arises upon considering the strictly massless limit², in which the neutrino mass eigenstates reduce to two-component Weyl fermions and helicity coincides exactly with chirality. In this limit, the physical neutrino exists only as the left-handed projection $\nu_{\alpha,\beta(i,j)L}$, while the corresponding antineutrino is purely right-handed, and no Lorentz transformation can connect the two. Since the bilinears $\sigma_{\mu\nu}q^\nu$ and $\sigma_{\mu\nu}q^\nu\gamma_5$ appearing in (3.26) are odd under chirality and therefore connect left- to right-handed spinors, their matrix elements between same-chirality Weyl states vanish identically, namely $\overline{\nu_{\alpha(i)L}} \sigma_{\mu\nu}q^\nu \nu_{\beta(j)L} = \overline{\nu_{\alpha(i)L}} \sigma_{\mu\nu}q^\nu \gamma_5 \nu_{\beta(j)L} = 0$, so that $\tilde{f}_M^{\alpha\beta}(\mathbb{f}_M^{ij})$ and $\tilde{f}_E^{\alpha\beta}(\mathbb{f}_E^{ij})$ are forced to vanish in the massless limit. The vertex function correspondingly collapses to

$$\tilde{\Lambda}_\mu^{\alpha\beta}(q) \Big|_{m_\nu=0} = (\gamma_\mu - q_\mu \not{q}/q^2) \left(\tilde{f}_Q^{\alpha\beta}(q^2) - \tilde{f}_A^{\alpha\beta}(q^2) q^2 \gamma_5 \right), \quad (3.31)$$

leaving the charge and anapole form factors as the only surviving electromagnetic structures available to a Weyl neutrino, consistent with the well-known statement that the magnetic and electric dipole moments of a Dirac neutrino are proportional to its mass and hence vanish identically in this limit. Moreover, since the one-loop SM diagrams contributing to the vertex function in the massless limit involve only same-flavor charged lepton propagators running in the loop, the vertex function is diagonal in the flavor basis: $\tilde{\Lambda}_\mu^{\alpha\beta} = 0$ and $\tilde{f}_{Q,A}^{\alpha\beta} = 0$ for $\alpha \neq \beta$ and each of the diagonal entries correspond to electromagnetic properties specific to each neutrino flavor.

3.3. Neutrino Charge Radius

Even if the neutrino is of neutral net charge, its charge form factor $f_Q(q^2)$ can still hold information as to its interaction with other charged particles. In the "Briet frame",

²We will show in later sections that this is consistent with the SM and a good approximation at the energy thresholds considered in this work.

defined by the time coordinate of four-momentum transfer such that $q_0 = 0$, $f_Q(q^2)$ only depends on $|\vec{q}| = \sqrt{-q^2}$ and can be interpreted as the Fourier transform of a spherically symmetric charge distribution $\rho(r)$ where $r = |\vec{r}|$.

$$f_Q(q^2) = \int \rho(r) e^{-i\vec{q}\cdot\vec{r}} d^3\vec{r} = \int \rho(r) \frac{\sin(|\vec{q}|r)}{|\vec{q}|r} d^3\vec{r}. \quad (3.32)$$

Deriving with respect to $q^2 = -|\vec{q}|^2$,

$$\frac{df_Q}{dq^2} = \int \rho(r) \frac{\sin(|\vec{q}|r) - |\vec{q}| \cos(|\vec{q}|r)}{2q^3 r} d^3\vec{r}, \quad (3.33)$$

and thus taking

$$\lim_{q^2 \rightarrow 0} \frac{df_Q}{dq^2} = \int \rho(r) \frac{r^2}{6} d^3\vec{r} = \frac{\langle r^2 \rangle}{6}. \quad (3.34)$$

Therefore, the average of the squared radius of the neutrino charge distribution is given by

$$\langle r^2 \rangle = 6 \left. \frac{df_Q}{dq^2} \right|_{q^2=0}. \quad (3.35)$$

Expanding $f_Q(q^2)$ around $q^2 = 0$ in a Taylor series yields

$$\begin{aligned} f_Q(q^2)|_{q \approx 0} &\approx f_Q(0) + \left. \frac{df_Q}{dq^2} \right|_{q^2=0} q^2 \\ &\approx \frac{\langle r^2 \rangle}{6} q^2, \end{aligned} \quad (3.36)$$

and substituting (3.36) into a matrix element of a conserved current with a vertex function of the form in (3.31) at the same limit gives

$$\begin{aligned} \overline{v}_L(p_f) \Lambda_\mu^{\text{SM}}(q) v_L(p_i) \Big|_{q^2 \approx 0} &= \overline{v}_L(p_f) (\gamma_\mu - g_\mu \not{q} + q^2) (f_Q(q^2) + f_A(q^2) q^2 \gamma_5) v_L(p_i), \\ &\approx \overline{v}_L(p_f) (\gamma_\mu) \left(\frac{\langle r^2 \rangle}{6} - \mathfrak{a} \right) q^2 v_L(p_i), \end{aligned} \quad (3.37)$$

utilizing $\gamma_5 v_L = -v_L$, $\overline{v}_L(p_f) \not{q} v_L(p_i) = 0$, and $f_A(0) = \mathfrak{a}$. Because of the form of Λ_μ^{SM} implied in (3.37) it's actually a matter of convention between authors what gets defined as the NCR since defining the total neutrino form factor $F_\nu(q^2)$ such that

$$\Lambda_\mu^{\text{SM}}(q) = \gamma_\mu F_\nu(q^2), \quad (3.38)$$

where $F_\nu(q^2) = f_Q(q^2) + q^2 f_A(q^2)$ acts as such that a SM $\nu\nu\gamma$ effective coupling cannot distinguish between an anapole and a charge form factor contribution at the $q \rightarrow 0$ limit and thus $F_\nu(q^2) \Big|_{q^2 \rightarrow 0} = f_Q(q^2) \Big|_{q^2 \rightarrow 0} \implies \mathfrak{a} = -\langle r^2 \rangle/6$ for ultra-relativistic left-hand neutrinos. This results in a redefining of an average square charge radius not in terms of f_Q , but rather F_ν as a whole which we will denote $\langle r^2 \rangle^{\text{SM}}$

$$\Lambda_\mu^{\text{SM}}(q) = \gamma_\mu F_\nu(q^2) \simeq \gamma_\mu q^2 \frac{\langle r^2 \rangle^{\text{SM}}}{6} \simeq \gamma_\mu q^2 \frac{\langle r^2 \rangle}{3}. \quad (3.39)$$

Different authors report either $\langle r^2 \rangle$ or $\langle r^2 \rangle^{\text{SM}}$ and thus the literature contains differences of factors of 2 for the NCR as seen from (3.39). The usual convention for contemporary neutrino phenomenology, and the one that will be used in this work, is to define

$$\langle r^2 \rangle^{\text{SM}} = +6 \frac{dF_\nu(q^2)}{dq^2} \Big|_{q=0}, \quad (3.40)$$

and report $\langle r^2 \rangle^{\text{SM}}$ as the NCR.

3.4. Corrections to the $\nu\nu\gamma$ vertex

The SM prediction for $\langle r^2 \rangle_{\alpha\alpha} = \langle r_{\nu_\alpha}^2 \rangle$ as defined by

$$\langle r_{\nu_\alpha}^2 \rangle = +6 \frac{d\tilde{f}^{\alpha\alpha}}{dq^2} \Big|_{q^2=0} \quad (3.41)$$

arises at one loop from radiative corrections to the $\nu\nu\gamma$ vertex. Within the pinch technique (PT) framework of [25], one constructs a gauge-invariant, process-independent effective vertex $\hat{\Gamma}_{\gamma\nu\bar{\nu}}^\mu$ by systematically extracting all longitudinal momenta from the conventional vertex, box, and self-energy diagrams and reassigning the resulting propagator-like pieces via the elementary Ward identities of the theory. The diagrams that contribute to $\hat{\Gamma}_{\gamma\nu\bar{\nu}}^\mu$ after this rearrangement are those in which a virtual W boson carries the photon coupling while the associated charged lepton ℓ_α runs in the loop; the remaining diagrams either cancel in the PT procedure or are absorbed into the effective self-energies $\hat{\Pi}_{gZ}^{\mu\nu}$ and $\hat{\Pi}_{ZZ}^{\mu\nu}$, which are instead interpreted as process-independent corrections to the effective weak mixing angle [25]. Crucially, the W vertex is diagonal in flavor, it couples ν_α exclusively to ℓ_α , so the loop integral depends only on m_{ℓ_α} and M_W , and the resulting charge form factor $\tilde{f}_Q^\alpha(q^2)$ is diagonal in the flavor basis. The Dirac form factor extracted from $\hat{\Gamma}_{\gamma\nu\bar{\nu}}^\mu$ then yields, in the limit $q^2 \rightarrow 0^3$ [25],

$$\langle r_{\nu_\alpha}^2 \rangle = \frac{\langle r_{\nu_\alpha}^2 \rangle^{\text{SM}}}{2} = -\frac{G_F}{4\sqrt{2}\pi^2} \left[3 - 2 \ln \frac{m_{\ell_\alpha}^2}{M_W^2} \right]. \quad (3.42)$$

which is finite, gauge-independent, and depends only on the flavor index α through the charged lepton mass m_α .

Numerically, the values of NCR prediction for each known neutrino flavour in the SM are

$$\langle r_{\nu_e}^2 \rangle^{\text{SM}} = -0.83 \times 10^{-32} \text{ cm}^2, \quad (3.43)$$

$$\langle r_{\nu_\mu}^2 \rangle^{\text{SM}} = -0.48 \times 10^{-32} \text{ cm}^2, \quad (3.44)$$

$$\langle r_{\nu_\tau}^2 \rangle^{\text{SM}} = -0.30 \times 10^{-32} \text{ cm}^2. \quad (3.45)$$

³utilizing the definition in (3.40)

3.5. NCR as a radiative correction of Electroweak Mixing

The expression (3.39) can be shown to be a correction that adds coherently to the tree-level weak neutral current amplitude, since both couple left-handed neutrinos to the same conserved vector current of the target. The tree-level weak neutral current vertex involves the product $G_F g_V$, where g_V is the vector coupling of the target fermion to the Z , and the propagator structure of the weak amplitude at low energies effectively replaces $q^2/M_Z^2 \rightarrow 0$ at leading order. The electromagnetic amplitude, by contrast, carries an explicit factor of q^2 from $f_Q(q^2)$ which cancels the photon propagator $1/q^2$, leaving a finite, q^2 -independent residue at leading order. The interference of this residue with the weak amplitude then shifts the effective vector coupling as

$$g_V \rightarrow g_V + \frac{2\sqrt{2}\pi\alpha}{3G_F} \langle r^2 \rangle, \quad (3.46)$$

where α is the fine-structure constant [26, 27]. Since $g_V^p = T_f^p - 2Q^p \sin^2 \theta_W = \frac{1}{2} - 2 \sin^2 \theta_W$ for the proton, this shift can equivalently be absorbed into a renormalization of the weak mixing angle,

$$\sin^2 \theta_W \rightarrow \sin^2 \theta_W \left(1 + \frac{1}{3} M_W^2 \langle r^2 \rangle \right), \quad (3.47)$$

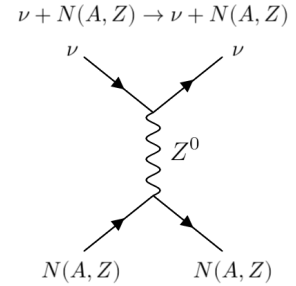
which is the form most commonly quoted in the literature [26, 27, 28]. The two shifts (3.46) and (3.47) are equivalent at leading order in α and their derivation makes transparent why the NCR promotes $\sin^2 \theta_W$ into a running coupling: the charge form factor $f_Q(q^2)$ carries the full q^2 dependence of the electromagnetic loop, and once this is included in the effective amplitude the vector coupling acquires a momentum-transfer dependence that is absent at tree level. In this sense the measurement of the NCR via CE ν NS is precisely a measurement of the running of the weak mixing angle from the Z pole down to the momentum transfers accessible at reactor and spallation neutron source experiments, wherein the SM prediction (??) fixes the expected magnitude and sign of the effect.

4. Coherent Elastic Neutrino-Nucleus Scattering

This chapter introduces the main interaction by which the sensitivities of neutrino detectors to the SM parameters are determined, describing a neutral current interaction along with its cross section, and showing how, in conjunction with a model of the antineutrino flux from a reactor source, the expected number of events for a broadly-described experimental design can be calculated; this number will be used in later chapters to test how different values of the SM parameters influence it.

4.1. Introduction

Coherent Elastic Neutrino-Nucleus Scattering (CE ν NS) is a neutral current weak interaction within the SM originally proposed in 1979 [29] and measured successfully in 2017 [30], wherein low-energy neutrinos [$E_\nu \sim \text{MeV}$] collide with atomic nuclei as coherent bodies rather than their constituents, imparting on them a small recoil energy. These interactions are possible with neutrino energies up to 50 MeV with medium-sized nuclei, since their main condition consists of the momentum transfer, Q , being negligible in comparison with the nuclear radius, R ($QR \ll 1$) [31]. The effective Feynman diagram for these interactions is shown in Fig. ??.



The calculation of its differential cross-section in the low-energy limit consists of setting an effective Lagrangian density given by:

$$\mathcal{L}_{\text{eff}} = \frac{G}{\sqrt{2}} j^{(\nu)\mu} \mathcal{J}_\mu, \quad (4.1)$$

where $j^{(\nu)\mu}$ denotes a neutral neutrino current and $\mathcal{J}_\mu$ a hadronic current from which the nuclear mass distribution terms originate.

4.2. Cross Section

The low-energy differential cross-section in terms of the nuclear recoil, E_R , is given by [32]:

$$\frac{d\sigma}{dT}(E_\nu, T) = \frac{G_F^2 M}{\pi} Q_W^2 \left(1 - \frac{ME_R}{2E_\nu} \right), \quad (4.2)$$

where M is the target nuclear mass and Q_W^2 is the vector nuclear weak charge given in terms of the proton (p) and neutron (n) weak couplings:

$$Q_W^2 = [Z g_V^p F_p(q^2) + N g_V^n F_n(q^2)]^2, \quad (4.3)$$

where g_V^p and g_V^n are given by (2.42) in the specific cases for the proton and neutron respectively:

$$g_V^p = T_p^3 - 2 \sin^2(\theta_W) Q_p = \frac{1}{2} - 2 \sin^2(\theta_W), \quad (4.4)$$

$$g_V^n = T_n^3 - 2 \sin^2(\theta_W) Q_n = -\frac{1}{2}, \quad (4.5)$$

and $F_{n,p}(Q^2)$ are the neutron and proton nuclear form factors, dependent on the momentum transfer $q^2 = 2MT$. In Eqs. (4.3), (4.4), and (4.5), Z is the atomic number, $N = A - Z$ is the neutron number, and A is the mass (nucleon) number of the target nucleus. It is worth noting that, given the weak mixing angle $\sin^2(\theta_W) \approx 0.22290(30) \approx \frac{1}{4}$ [33], Q_W^2 (and by extension $\frac{d\sigma}{dT}$) is mostly $\propto N^2$.

4.3. Neutrino Sources

Part of the calculation of the expected number of CE ν NS events is the energy-dependent flux of incoming (anti)neutrinos. This work focuses mainly on a model for electron-antineutrinos known as the Huber-Mueller model.

4.3.1 Nuclear Reactors

Ever since their discovery in 1959 [2], reactor neutrino sources have been fundamental for neutrino physics. In recent times they have still been utilized in experiments such as the measurement of Δm_{21} in KamLAND [34] and of the mixing angle θ_{13} in Daya Bay [35]. The main inspiration for the present work comes from two papers, one by L. J. Flores et al. [36] and another by E. Alfonso-Pita et al. [37], wherein a 10 kg scintillating liquid Argon (LAr) bubble chamber detector is placed at a distance of ~ 3 m from a reactor running at ~ 1 MW_{th} thermal power, based on a TRIGA Mark III nuclear reactor at the National Institute of Nuclear Research (ININ) facilities on the Mexico-Toluca highway near Mexico City.

Nuclear reactors commonly operate based on Uranium and Plutonium, whose fission produces a large quantity of neutron-rich remnant nuclei that subsequently undergo beta decay. The resulting antineutrino flux for ^{235}U , ^{239}Pu , and ^{241}Pu has been calculated by the Institut Laue-Langevin (ILL) in the 1980s [38, 39]. Coupled with an *ab initio* method for the estimation of the ^{238}U spectrum as seen in [40], the following model is obtained for the total electron antineutrino spectrum, known as the Huber-Mueller model [41]:

$$\frac{d\Phi}{dE_\nu}(E_\nu) = \frac{P}{4\pi d^2} \sum_k \frac{f_k}{\epsilon_k} \exp\left(\sum_{p=1}^6 \alpha_{kp} E_\nu^{p-1}\right), \quad (4.6)$$

where P is the thermal power of the reactor, d is the reactor-detector distance, $k \in \{^{235}\text{U}, ^{239}\text{Pu}, ^{241}\text{Pu}, ^{238}\text{U}\}$ is an index that runs through the contribution of the isotopes involved, ϵ_k is the energy per fission of each isotope as published in [42], f_k is the fission fraction of each isotope, and α_{kp} are the experimentally fitted coefficients shown in Table 4.1. In Fig. 4.1 the individual isotope contributions are plotted against each other in arbitrary units, and Fig. 4.2 shows the total flux given the fission fractions: 55% of ^{235}U , 32% of ^{239}Pu , 6% of ^{241}Pu , and 7% of ^{238}U , in accordance with [43].

	^{235}U	^{238}U	^{239}Pu	^{241}Pu
α_1	3.217	$4.833 \cdot 10^{-1}$	6.413	3.251
α_2	-3.111	$1.927 \cdot 10^{-1}$	-7.432	-3.204
α_3	1.395	$-1.283 \cdot 10^{-1}$	3.535	1.428
α_4	$-3.690 \cdot 10^{-1}$	$-6.762 \cdot 10^{-3}$	$-8.820 \cdot 10^{-1}$	$-3.675 \cdot 10^{-1}$
α_5	$4.445 \cdot 10^{-2}$	$2.233 \cdot 10^{-3}$	$1.025 \cdot 10^{-1}$	$4.254 \cdot 10^{-2}$
α_6	$-2.053 \cdot 10^{-3}$	$-1.536 \cdot 10^{-4}$	$-4.550 \cdot 10^{-3}$	$-1.869 \cdot 10^{-3}$

Table 4.1: Experimentally fitted coefficients α_{kp} as published in [41].

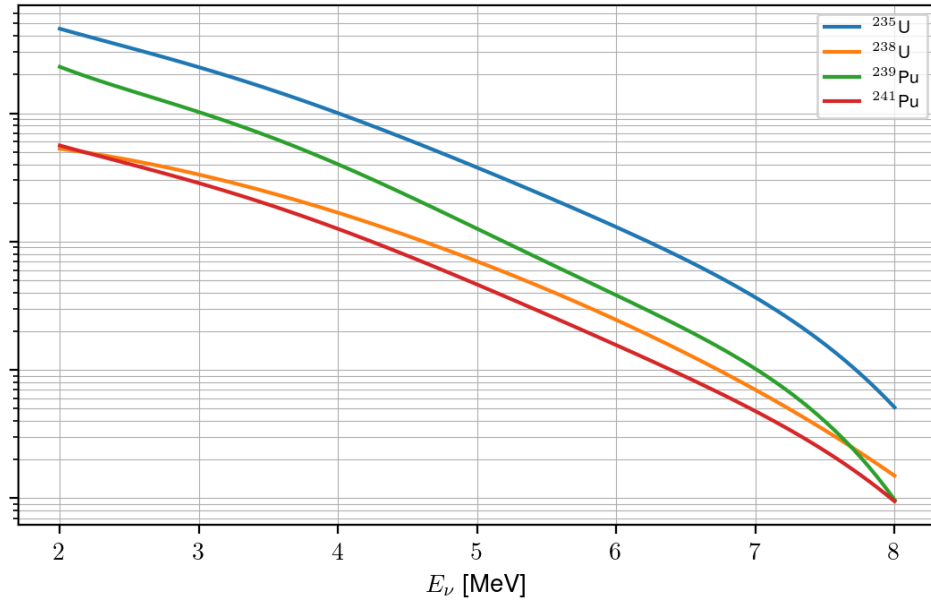


Figure 4.1: Isotope contributions to total antineutrino flux spectrum.

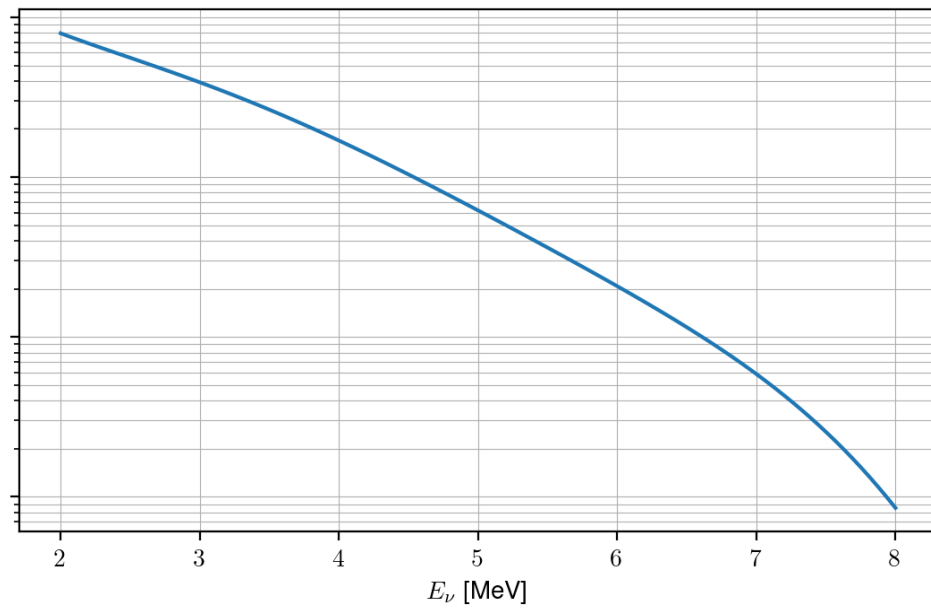


Figure 4.2: Total antineutrino flux spectrum.

4.4. Expected Number of Events

Ignoring reactor resolution and efficiency, the expected number of events of a detector of mass M over an exposure time t is given by

$$N = t \frac{M_{\text{Detector}}}{M} \int_{E_{R \text{ th}}}^{E_{R \text{ max}}} dE_R \int_{E_{\nu, \text{ min}}(E_R)}^{E_{\nu, \text{ max}}} dE_\nu \frac{d\Phi}{dE_\nu}(E_\nu) \left(\frac{d\sigma}{dE_R} \right), \quad (4.7)$$

where $E_{R \text{ th}}$ a threshold nuclear recoil energy, $E_{R \text{ max}}$ a kinematically determined maximum recoil energy, $E_{\nu, \text{ max}} \sim 9.5$ MeV the maximum incident neutrino energy given by the neutrino spectrum [39] and $E_{\nu, \text{ min}}(E_R) \approx \sqrt{ME_R/2}$ the minimum incident neutrino energy required to produce a given E_R recoil energy

$$E_{\nu, \text{ min}}(E_R) = \frac{E_R}{2} \left(1 + \sqrt{1 + \frac{2M}{E_R}} \right). \quad (4.8)$$

5. SBC-ININ-LAr

5.1. Results

6. Summary and conclusions

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